Wasting Away In Hidden Subgroup Ville



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To: Dave Bacon
From: Patrick Hayden

> Tuesday, December 13: IQI Seminar
> "Wasting away in hidden subgroup ville"

Oh yes. I'm afraid we've seen this before. Obsessive-compulsive manic consumptive research disorder. There is no cure, but it is treatable with alternating doses of unwarranted optimism and gaping-abyss pessimism.

In the Beginning There Was…Factoring

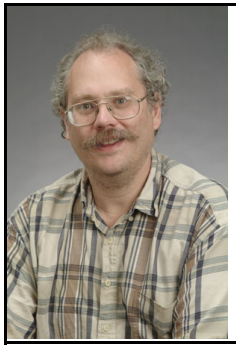
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472772146107435302536223071973048224632914695302097116459852171130520711256363590397527



Peter Shor's 1994 discovery

quantum computers can efficiently factor and compute the discrete logarithm

Reaction #1
Run For the Hills



“Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world’s most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security.” -Simon Singh

Reaction #2
Not Even Wrong



2. Supernatural performance myth

The belief in the extraordinary performance of quantum computer is based mainly on the claim by Shor's quantum factorization of a large integer **n** algorithm, to be faster than **O(n)**.

But, it is obvious that it is much slower than **O(n)**, and is in fact the worst performance disaster in computational history.

http://mywebpages.comcast.net/mycommon/Quantum_Computing_Myths.PDF

Reaction #3
Dreams of Future Algorithms



“For my part I know nothing with any certainty, but the sight of the stars makes me dream.” – Vincent van Gogh

“I wasted time, and now doth time waste me”

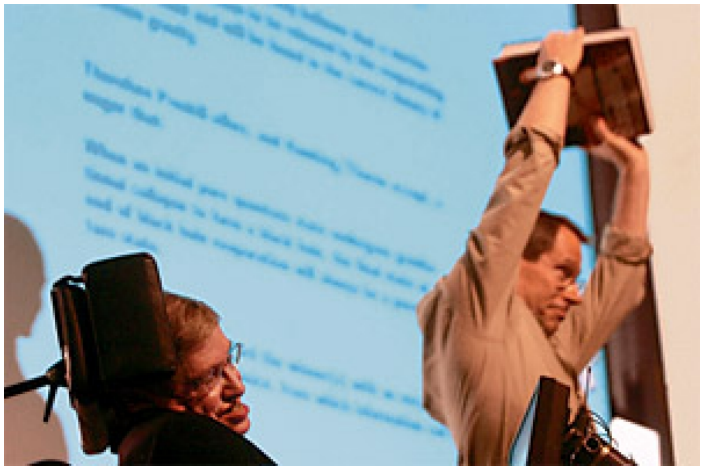
Caltech SURF, 1996

Quantum Computation and NP-Completeness

Dave Bacon
Summer Undergraduate Research Fellowship Paper
Advisors: Nicholas Cerf and Chris Adami
California Institute of Technology

“That was a hard problem.” – John Preskill





Revolutions

Relativity Revolutions

1905: Special Relativity
1915: General Relativity

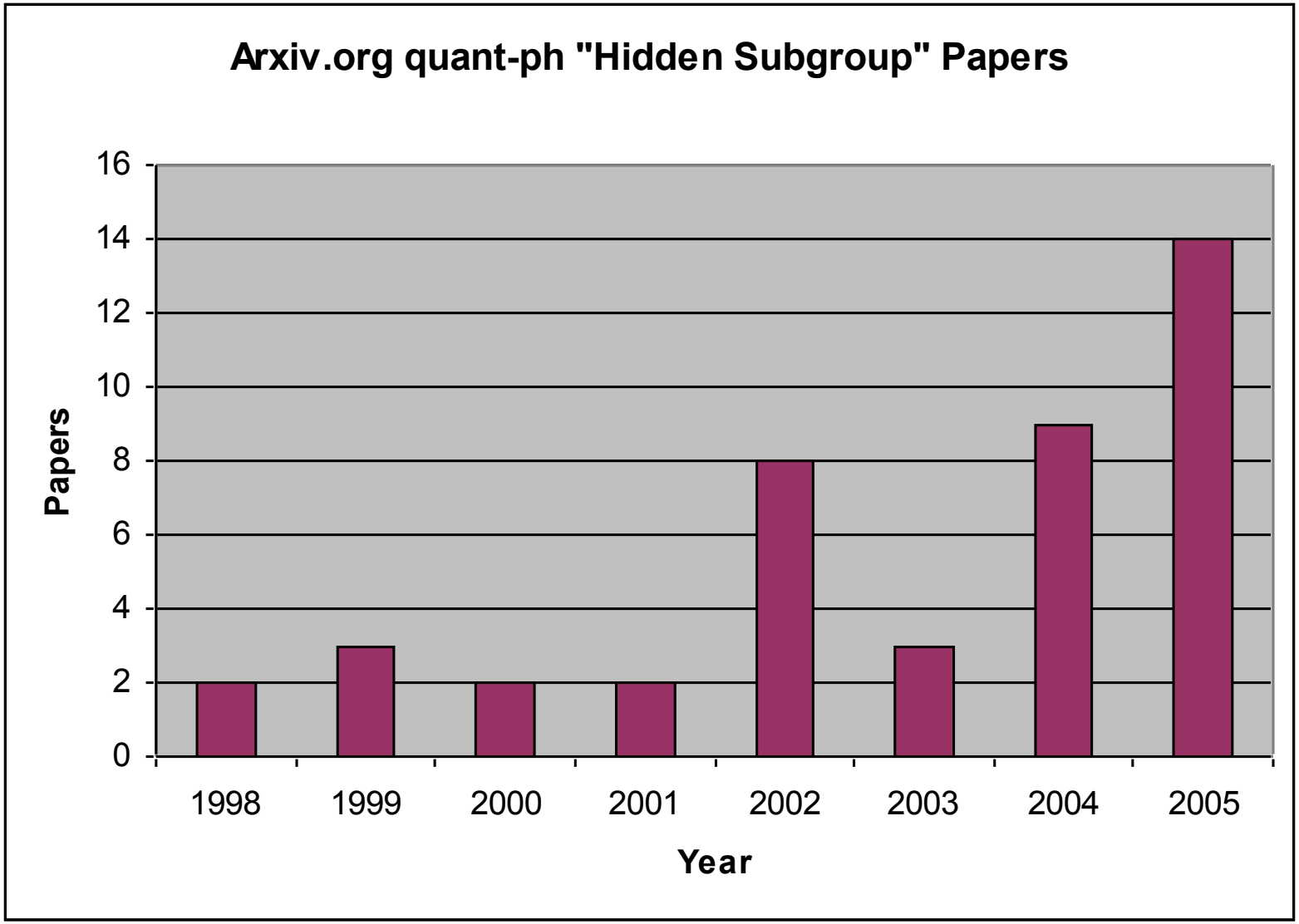
String Theory Revolutions

1984: Green and Schwarz anomaly cancellations
1994: Dualities between string theories

The Time Unit of Revolutions is the Decade

This means we are OVERDUE for a revolution in quantum computing!

Hidden Subgroup Problem



Today

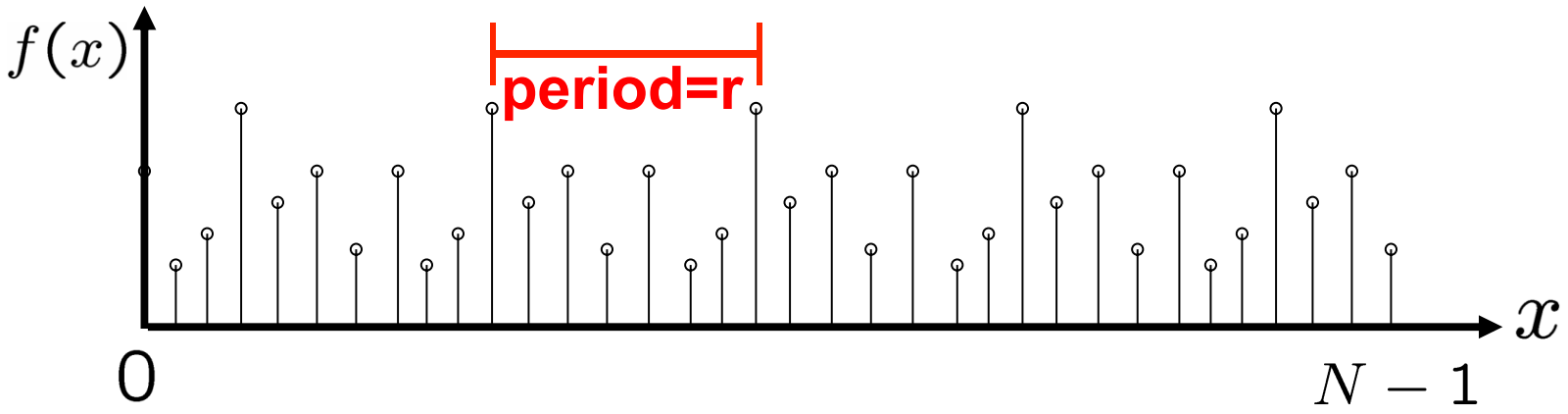
Where are we going? Why are we going there?



Structure of the Problem?

Shor's Factoring Algorithm

Factoring can be reduced to order finding
Order finding can be reduced to period finding.



Given query access to a function $f : \mathbb{Z}_N \to \mathbb{Z}_N$ with the promise

$$f(x) = f(y) \quad \text{iff} \quad x = y + kr \mod N$$
$$k \in \mathbb{Z}$$

Find the period r

Groups

Definition

Group: Set of elements $\mathcal{G}$ with binary multiplication rule $\mathcal{G} \times \mathcal{G} \to \mathcal{G}$

Associative: $g_1(g_2 g_3) = (g_1 g_2) g_3$

Identity $ge = g$

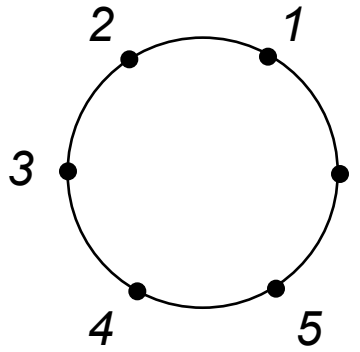
İnverse: $gg^{-1} = e$



Group $\mathbb{Z}_N$

elements: $\{0, 1, 2, \ldots, N-1\}$

group multiplication: $+$



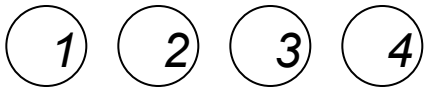
$\mathbb{Z}_6$

Symmetric Group $\mathcal{S}_n$

elements: permutations of n objects

$\{e, (12), (23), \ldots, (123), \ldots\}$

group multiplication: composition of permutations



$\mathcal{S}_4$

(23)

Shor's Hidden Subgroup

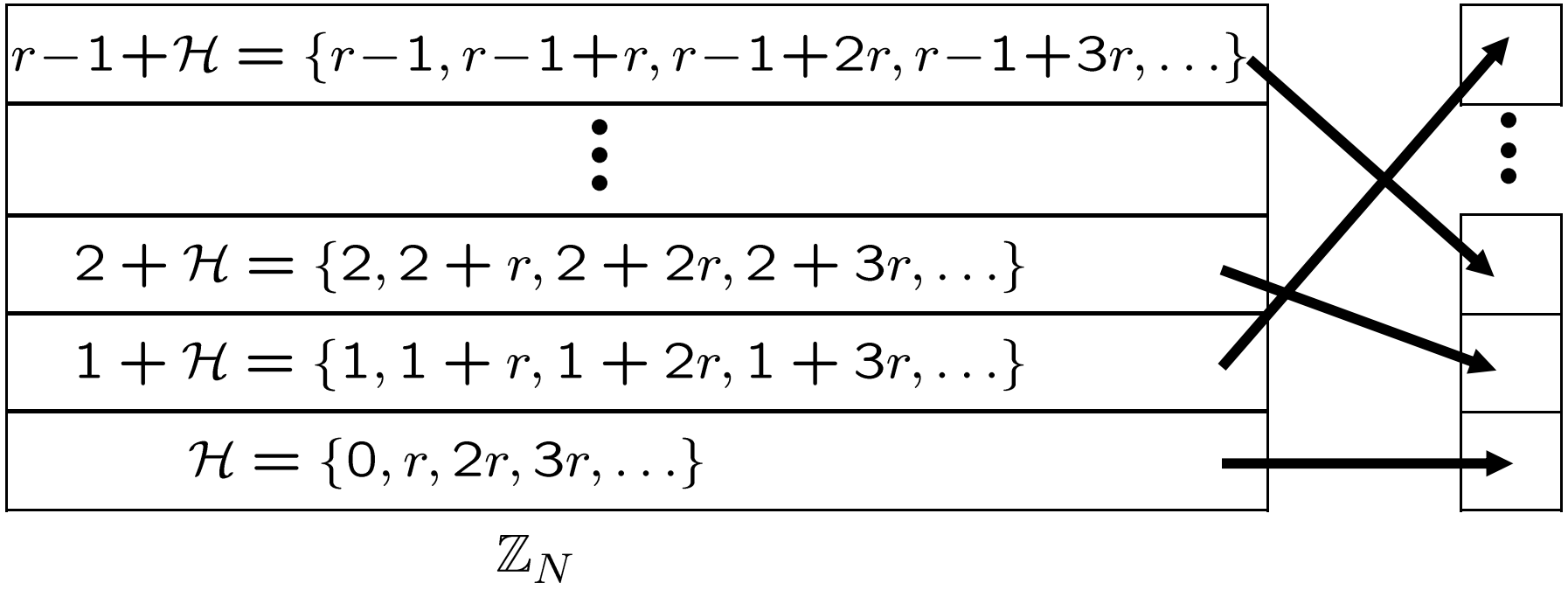
Given query access to a function $f : \mathbb{Z}_N \to \mathbb{Z}_N$ with the promise

$$f(x) = f(y) \quad \text{iff} \quad x = y + kr \bmod N, \quad k \in \mathbb{Z}$$

Subgroup $\mathcal{H} \subset \mathbb{Z}_N$

$$\mathcal{H} = \{0, r, 2r, 3r, \ldots\}$$

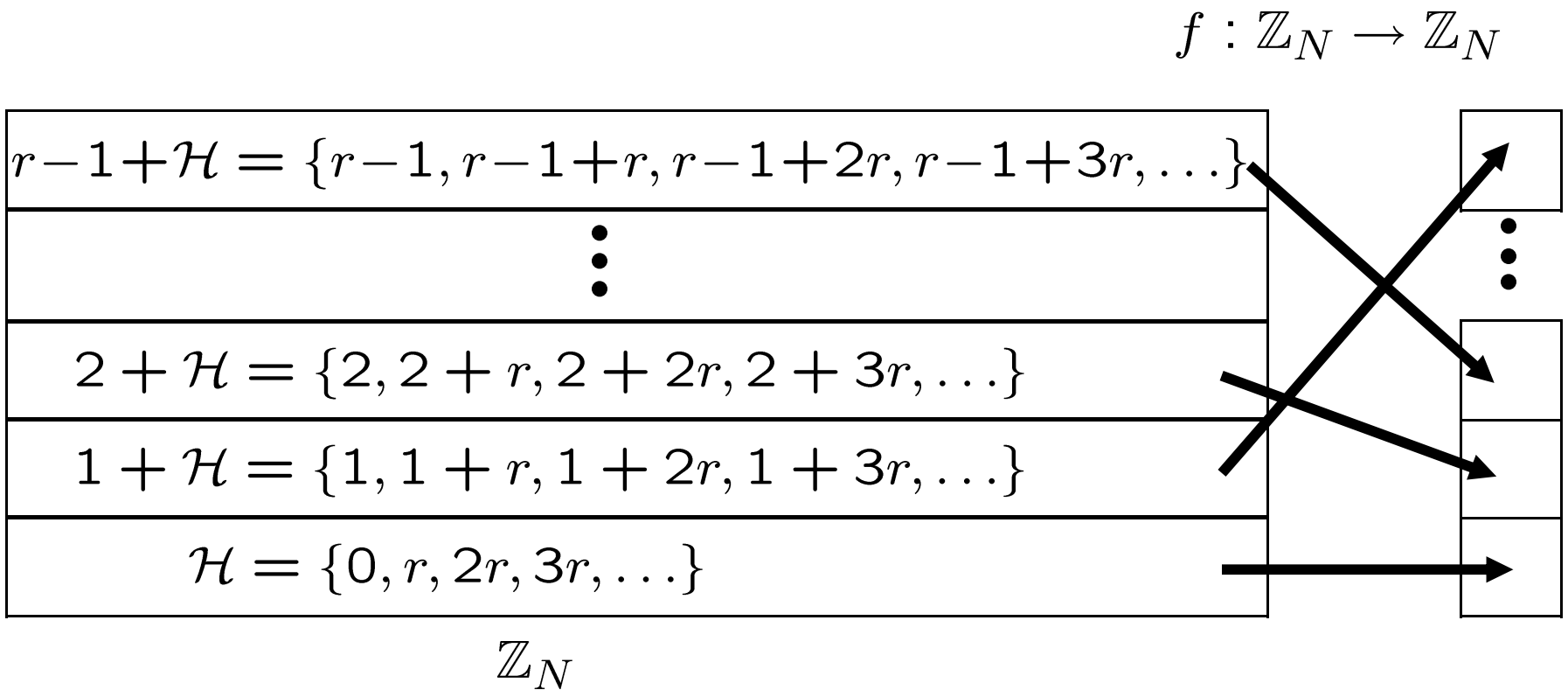
$$f : \mathbb{Z}_N \to \mathbb{Z}_N$$



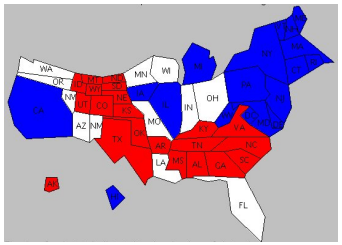
Shor's Hidden Subgroup

Given query access to a function $f : \mathbb{Z}_N \to \mathbb{Z}_N$ with the promise

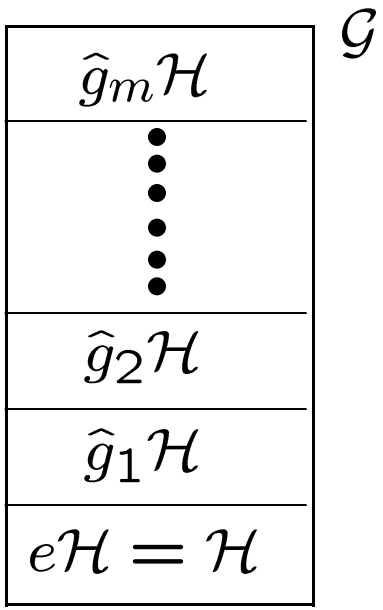
$f(x)$ is constant and distinct on cosets of $\mathcal{H}$



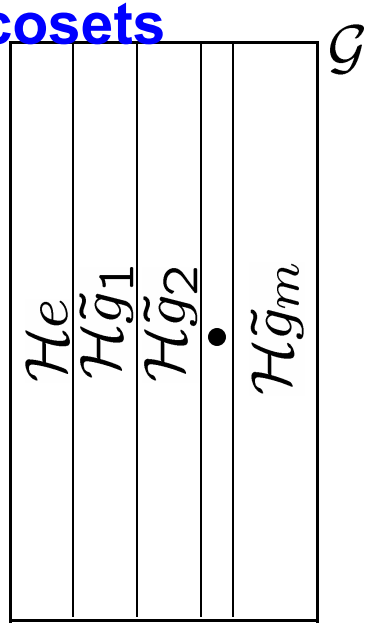
Cosets



Left cosets



Right cosets



$$g\mathcal{H} = \{gh | h \in \mathcal{H}\}$$

$$\mathcal{H}g = \{hg | h \in \mathcal{H}\}$$

Unless $\mathcal{H}$ is normal, then the left and right cosets do not coincide

Definition

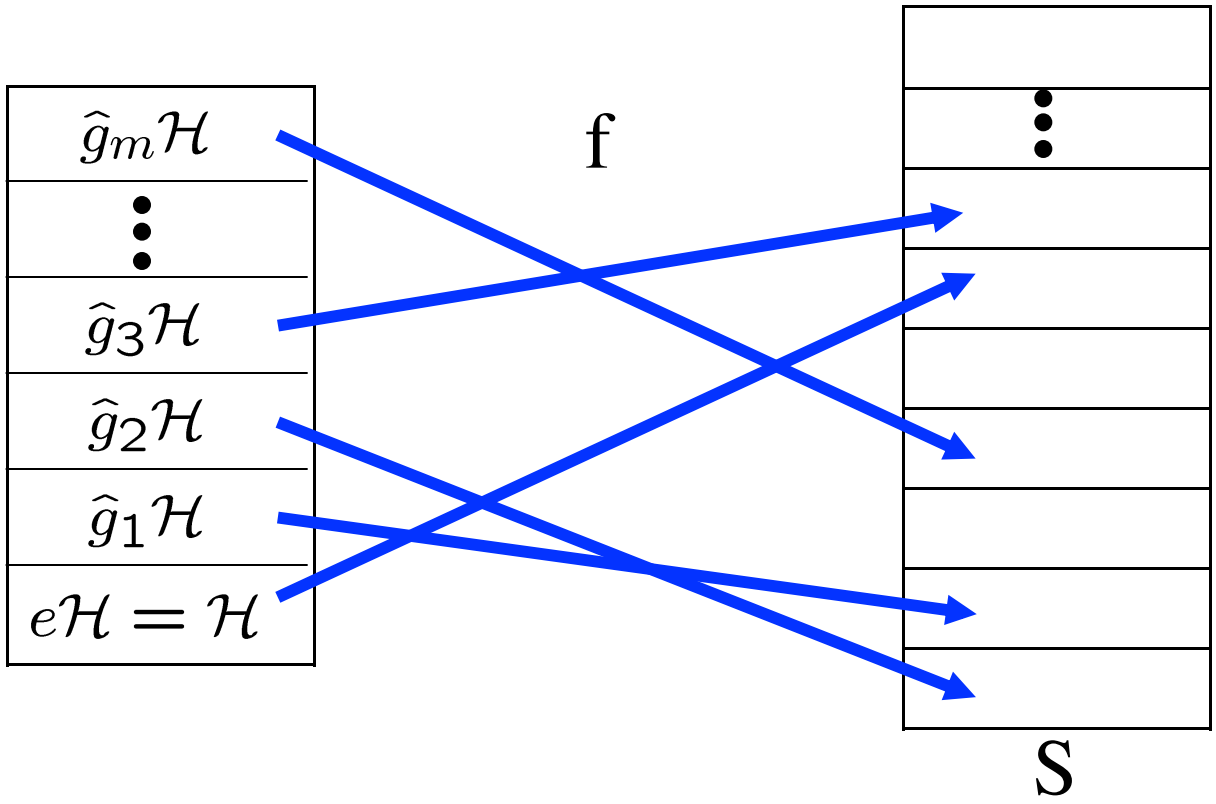
Normal subgroup: $ghg^{-1} \in \mathcal{H}, \forall h \in \mathcal{H}, g \in \mathcal{G}$

Hidden Subgroup Problem

Setup: Fix a group $\mathcal{G}$(known) and a subgroup $\mathcal{H} \subset \mathcal{G}$(unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$ We say that f hides $\mathcal{H}$

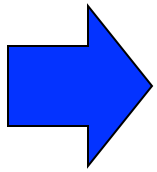
Problem: Find a generating set for $\mathcal{H}$ (identify $\mathcal{H}$).

Efficient: If runs in time polylog($|\mathcal{G}|$)



Why the HSP?

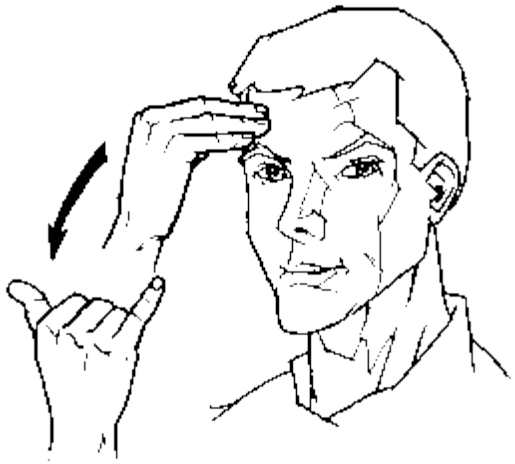
efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$)



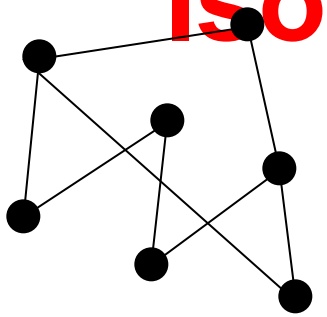
efficient algorithm for the graph isomorphism problem

efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$

efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

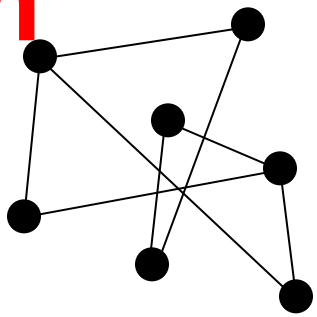


Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the V_2 vertices such that the edges are respected, $p(E_1) = E_2$?

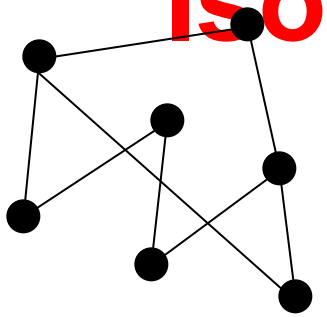
Mapping to the HSP:

$$f : \mathcal{S}_{2n} \rightarrow \mathbb{Z}_{\frac{n(n-1)}{2}} \times \mathbb{Z}_{\frac{n(n-1)}{2}}$$

$$f(\pi) = \pi(A_1 \times A_2)$$

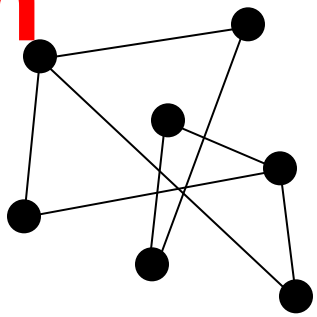
If graphs are not isomorphic, hidden subgroup will not mix the graphs. If they are isomorphic, they will mix the graphs.

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Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the V_2 vertices such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

$$f : \mathcal{S}_n \wr \mathcal{S}_2 \rightarrow \mathbb{Z}_{\frac{n(n-1)}{2}} \times \mathbb{Z}_{\frac{n(n-1)}{2}}$$

$$f((\pi_1, \pi_2, \tau)) = \tau(\pi_1(A_1) \times \pi_2(A_2))$$

Notice now we are only worried if hidden subgroup has a nontrivial τ

Conjugate Subgroups

Two subgroups, $\mathcal{H}_1$ and $\mathcal{H}_2$, are conjugate if there exists an element of the group $g \in \mathcal{G}$ such that

$$\mathcal{H}_1 = \{ghg^{-1}, h \in \mathcal{H}_2\}$$

This is an equivalence relation among the subgroups of $\mathcal{G}$

$$\mathcal{H}_i \equiv_{\mathcal{G}} \mathcal{H}_j$$

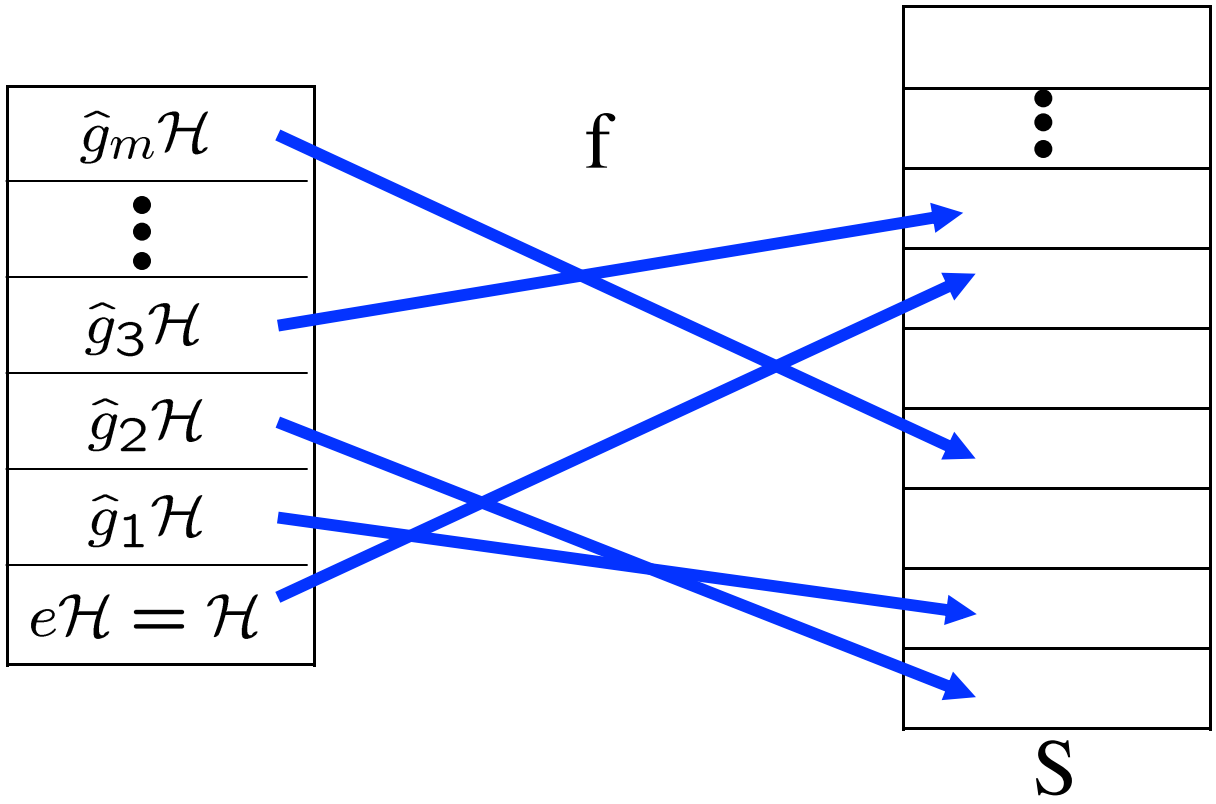
voy, vas, va, vamos,

Hidden Subgroup Conjugacy Problem

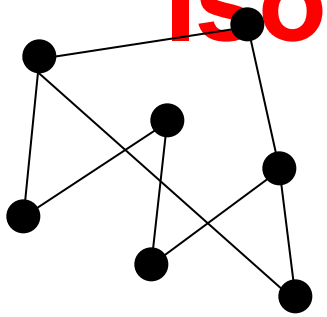
Setup: Fix a group $\mathcal{G}$(known) and a subgroup $\mathcal{H} \subset \mathcal{G}$(unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$ We say that f hides $\mathcal{H}$

Problem: identify the conjugate subgroup class of $\mathcal{H}$.

Efficient: If runs in time polylog($|\mathcal{G}|$)

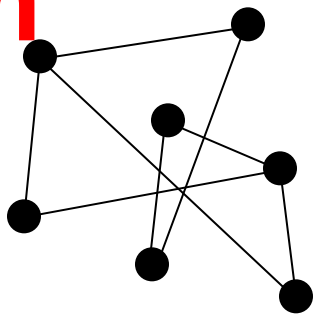


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Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSCP:

$$f : \mathcal{S}_n \wr \mathcal{S}_2 \rightarrow \mathbb{Z}_{\frac{n(n-1)}{2}} \times \mathbb{Z}_{\frac{n(n-1)}{2}}$$

$$f((\pi_1, \pi_2, \tau)) = \tau(\pi_1(A_1) \times \pi_2(A_2))$$

Notice now we are only worried if hidden subgroup conjugacy has a nontrivial τ

Relationships

How do the hidden subgroup problem (HSP) and the hidden subgroup conjugacy problem (HSCP) differ?

1. If group is abelian, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

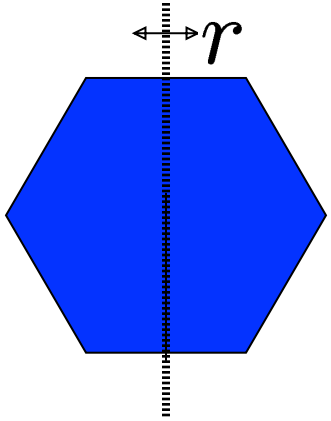
2. If subgroups are normal, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

These are two cases where we know quantum computers can efficiently solve the HSP.

Dihedral Group



Dihedral group $\mathcal{D}_N$

group of symmetries of a *N* sided regular polygon

$$\mathcal{D}_N = \{r^t s^k, t \in \mathbb{Z}_2, k \in \mathbb{Z}_N\}$$

$$r^2 = s^N = rsrs = e$$

s = rotation

r = reflection

Subgroups of $\mathcal{D}_N$

cyclic subgroups:

$$\mathcal{C}_{N/j} = \{e, s^j, s^{2j}, \ldots, s^{-j}\}$$

$$j \in \mathbb{Z}_N \quad j \in \mathbb{Z}_N \quad j|N$$

dihedral subgroups:

$$\mathcal{D}_{N/j,l} = \{e, s^j, \ldots, s^{-j}, rs^l, rs^{j+l}, \ldots, rs^{-j+l}\}$$

$$j \in \mathbb{Z}_N \quad j \in \mathbb{Z}_N \quad j|N \quad l \in \mathbb{Z}_N$$

Dihedral HSP

Theorem [Ettinger, Hoyer '98]

Sufficient to solve Dihedral HSP when subgroups are limited to

$$\mathcal{H}_l = \{e, rs^l\} \quad l \in \mathbb{Z}_N \quad \text{or} \quad \mathcal{T} = \{e\}$$

if N is odd, then the $\mathcal{H}_l = \{e, rs^l\}$ subgroups are all conjugate to each other:

$$s^{-k}\mathcal{H}_l s^k = \{e, rs^{l+2k}\}$$

solving the Dihedral HSCP does not immediately solve the Dihedral HSP…

Modification



Modify the function hiding the subgroup:

$$f : \mathcal{D}_N \to S$$
$$g : \mathcal{D}_N \to \{+1, -1\}$$
$$g(r^t s^k) = (-1)^k$$

left cosets:

$$\mathcal{H}_l = \{e, rs^l\}$$
$$s\mathcal{H}_l = \{s, rs^{l-1}\}$$
$$s^2\mathcal{H}_l = \{s, rs^{l-2}\}$$
$$\vdots$$

If l is even, then (f, g) is constant and distinct on left cosets of $\mathcal{H}_l$

If l is odd, then (f, g) is constant and distinct on left cosets of $\mathcal{T} = \{e\}$

Solving HSCP for new problem thus allows us to distinguish between odd and even subgroups

[Moore 05, personal communication]

Modification

Modify the function hiding the subgroup:

$$f : \mathcal{D}_N \to S$$

$$g_1 : \mathcal{D}_N \to \{+1, -1\}$$

$$g_2 : \mathcal{D}_N \to \{+1, -1\}$$

left cosets:

$$\mathcal{H}_l = \{e, rs^l\}$$
$$s\mathcal{H}_l = \{s, rs^{l-1}\}$$
$$s^2\mathcal{H}_l = \{s, rs^{l-2}\}$$
$$\vdots$$

Using more functions, we can determine 2nd least significant bit of ℓ.

Generalizing this allows us to use the Dihedral HSCP to solve the Dihedral HSP (they are polynomial time equivalent.)

[Moore 05, personal communication]

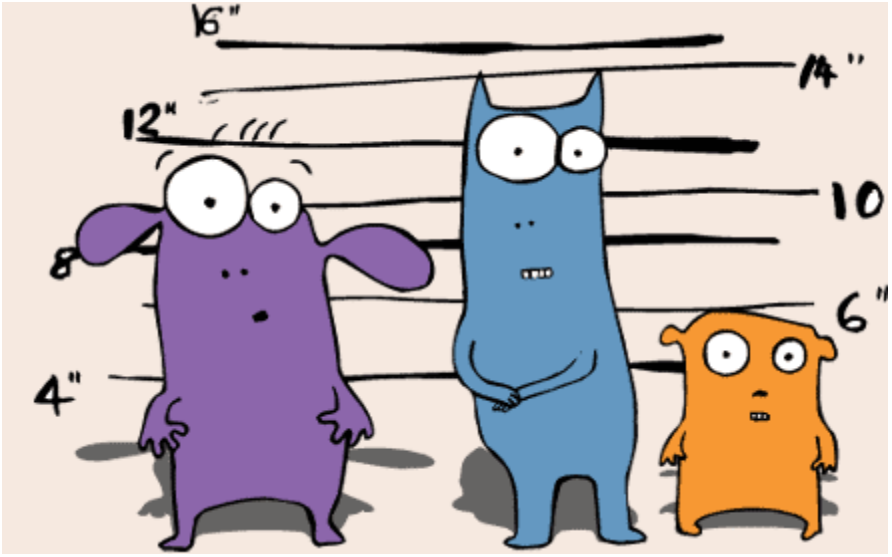
see also [Kuperberg 03]

HSP and HSCP

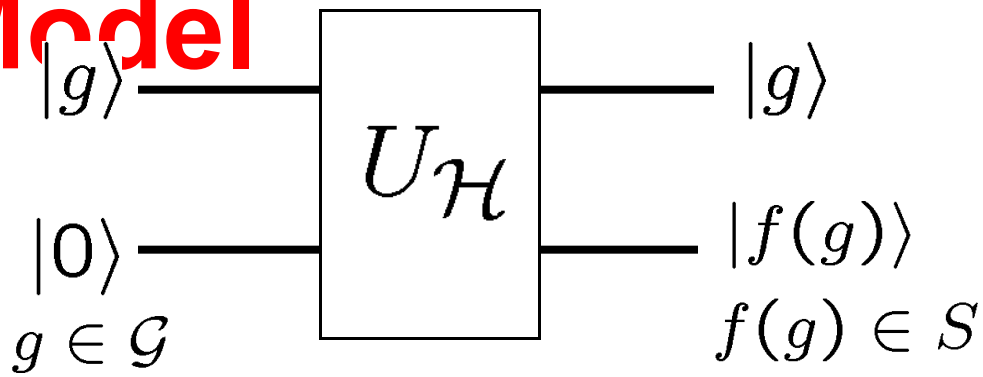
More generally, we can ask if this method can be extended to work for any group/set of subgroups?

Why the HSCP?

The HSCP has more “symmetry” than the HSP



Standard Query Model



query in superposition:

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

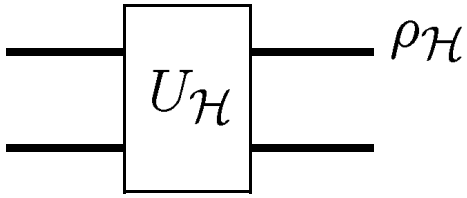
constant and distinct on different left cosets of

$$|g\mathcal{H}\rangle = \frac{1}{\sqrt{|\mathcal{H}|}} \sum_{h \in \mathcal{H}} |gh\rangle$$ random left coset state

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$ hidden subgroup state

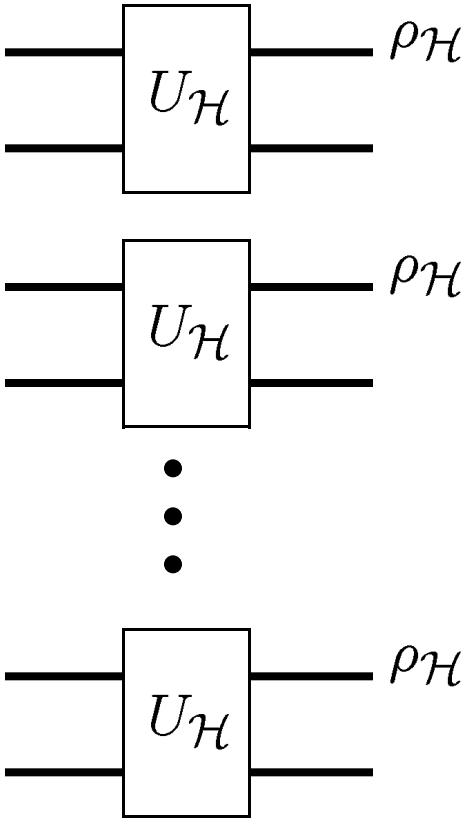
$\hat{\mathcal{G}} :=$ set of coset representatives

Multiple Queries



If we make a single measurement on the $\mathcal{H}$ HS state, can we distinguish different

NO



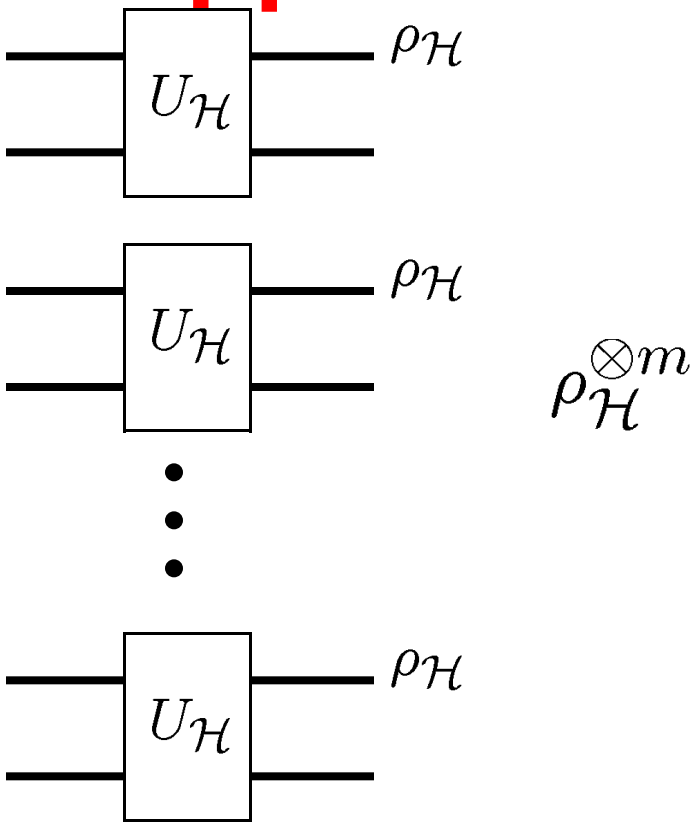
$\rho_{\mathcal{H}}^{\otimes m}$

We need multiple queries to distinguish different subgroups

$m = O(\log |\mathcal{G}|)$ sufficient

[Ettinger, Høyer, Knill 99]

Multiple Queries Approaches



$\rho_{\mathcal{H}}^{\otimes m}$

Existential

explicit measurement,
exponentially sized circuit

[Ettinger, Høyer, Knill 99]

Pretty Good Measurement

often optimal
efficient circuits for some groups,
but others appear hard

[Bacon, Childs, van Dam 05]

Imagine you have figured out how to solve the HSP
Describe to yourself the idea that makes your method work.

Symmetries



left regular representation

$$D_L(g)|g'\rangle = |gg'\rangle$$

right regular representation

$$D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$g, g' \in \mathcal{G}$$

they commute $D_L(g_1)D_R(g_2) = D_R(g_2)D_L(g_1)$

representation theory

$$D_L(g) \cong \bigoplus_{\mu} (D_\mu(g) \otimes I_{d_\mu})$$

$$D_R(g) \cong \bigoplus_{\mu} (I_{d_\mu} \otimes D_\mu(g))$$

sum over all irreps

The transform from the basis of group elements $|g\rangle$ to this block diagonal basis $|\mu, i, j\rangle$ is the Fourier transform over $\mathcal{G}$

$$Q_{\mathcal{G}} = \sum_{\mu} \sum_{i,j=1}^{d_\mu} \sum_{g \in \mathcal{G}} \sqrt{\frac{d_\mu}{|\mathcal{G}|}} [D_\mu(g)]_{i,j} |\mu, i, j\rangle\langle g|$$

Symmetries of HS States

left regular representation

$$D_L(g)|g'\rangle = |gg'\rangle$$

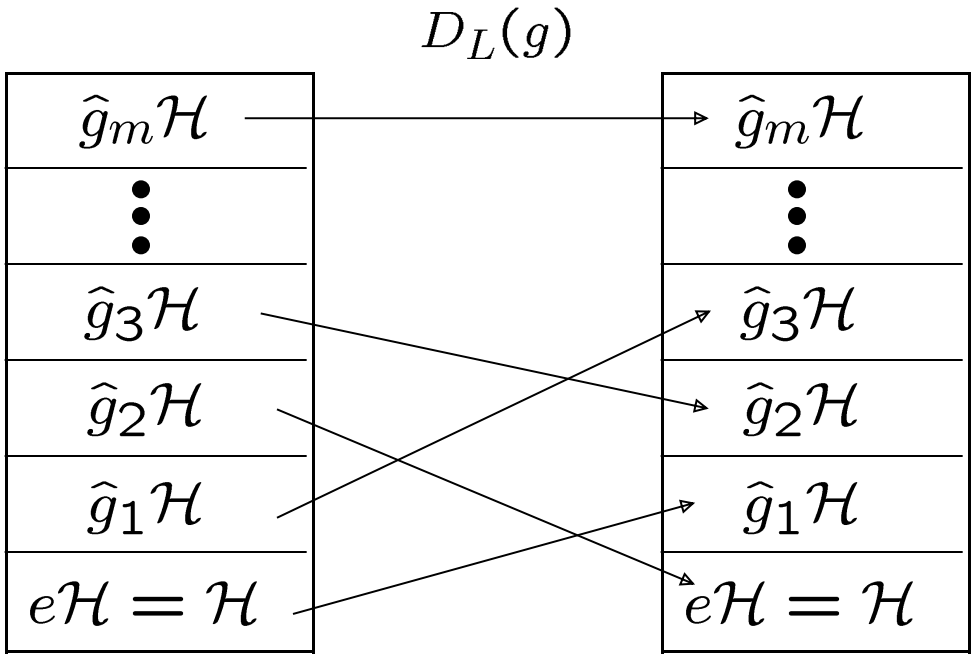
right regular representation

$$D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$g, g' \in \mathcal{G}$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$

$$D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$



Symmetries of HS States

left regular representation

$$D_L(g)|g'\rangle = |gg'\rangle$$

right regular representation

$$D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$g, g' \in \mathcal{G}$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$

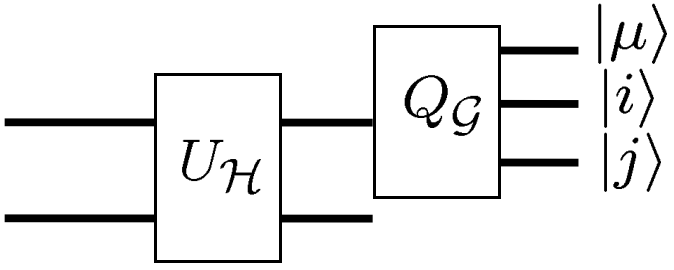
$$D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

via Schur's lemma this implies

$$\rho_{\mathcal{H}} = \frac{1}{|\mathcal{G}|} \sum_{h \in \mathcal{H}} D_R(h)$$

after applying the Fourier transform

$$\rho_{\mathcal{H}} \cong \bigoplus_{\mu} \left(I_{d_\mu} \otimes \frac{1}{|\mathcal{G}|} \sum_{h \in \mathcal{H}} D_\mu(h) \right)$$



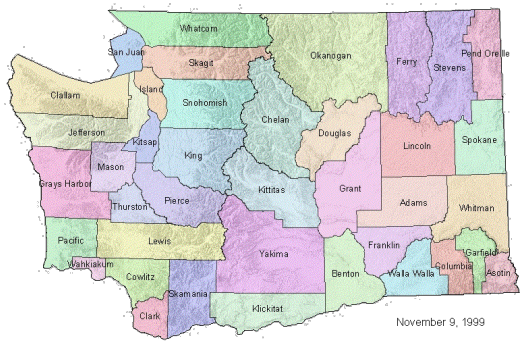
State Distinguishablity

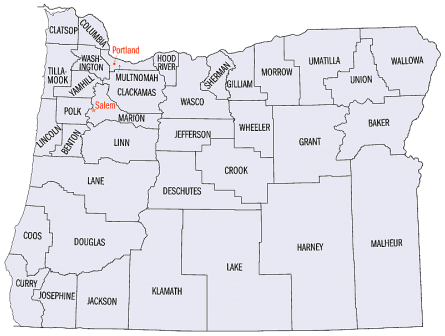
Given $\rho_{\mathcal{H}}$ with probability $p_{\mathcal{H}}$

Find measurement which maximizes probability of successfully identifying $\rho_{\mathcal{H}}$

[Ip 03, Bacon, Childs, van Dam 05, Moore, Russell 05, Bacon, Childs, van Dam 05, Childs, van Dam 05, Childs, Wocjan 05, …]

Often the measurement which maximizing this probability is the “pretty-good-measurement” (PGM)





State Distinguishablity

HSCP: we don't care about which subgroup, just it's conjugacy. New problem

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}}$$

Probability: $p_{[\mathcal{H}]}$

HSC state multiple queries:

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

Probability: $p_{[\mathcal{H}]}$

These states have been symmetricized!

Single HSC State

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}} = \frac{1}{|\mathcal{G}|^2} \sum_{g \in \mathcal{G}} \sum_{h \in \mathcal{H}} D_R(ghg^{-1})$$

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g) \rho_{\mathcal{H}} D_R(g^{-1})$$

$$\rho_{\mathcal{H}} \cong \bigoplus_{\mu} \left(I_{d_\mu} \otimes \frac{1}{|\mathcal{G}|} \sum_{h \in \mathcal{H}} D_\mu(h) \right)$$

Single HSC state is completely diagonalized (like Abelian!)

$$\rho_{[\mathcal{H}]} \cong \bigoplus_{\mu} c_\mu([\mathcal{H}]) \left(I_{d_\mu} \otimes I_{d_\mu} \right) \qquad c_\mu([\mathcal{H}]) = \frac{1}{d_\mu |\mathcal{G}|} \sum_{h \in \mathcal{H}} \chi_\mu(h)$$

Dihedral USCB

$$\rho_{[\mathcal{H}]} \cong \bigoplus_{\mu} c_\mu([\mathcal{H}]) \left(I_{d_\mu} \otimes I_{d_\mu}\right) \qquad c_\mu([\mathcal{H}]) = \frac{1}{d_\mu |\mathcal{G}|} \sum_{h \in \mathcal{H}} \chi_\mu(h)$$

Dihedral group $\mathcal{D}_N$

2 one dimensional irreps

$$\chi_T(r^t s^k) = 1 \qquad \chi_A(r^t s^k) = (-1)^t$$

$\frac{N-1}{2}$ two dimensional irreps $h \in \{1, 2, \ldots, \frac{N-1}{2}\}$

$$\chi_h(r^t s^k) = 2\delta_{t,0} \cos\left(\frac{2\pi h k}{N}\right)$$

	T	A	h
$c_\mu([\{e\}])$	$\frac{1}{2N}$	$\frac{1}{2N}$	$\frac{1}{2N}$
$c_\mu([\{e, r\}])$	$\frac{1}{N}$	0	$\frac{1}{2N}$

Optimal Measurement?



Pretty good measurement:

$$G_j = p_j T^{-1/2} \rho_j T^{-1/2} \qquad T = \sum_i p_i \rho_i$$

For Dihedral group HSCP, PGP is

$$G_{[\{e\}]} = [\frac{p}{2-p}] \oplus [1] \bigoplus_{h=1}^{\frac{N-1}{2}} (pI_2 \otimes I_2)$$

$$G_{[\{e,r\}]} = [\frac{2-2p}{2-p}] \oplus [0] \bigoplus_{h=1}^{\frac{N-1}{2}} ((1-p)I_2 \otimes I_2)$$

But this is not the optimal measurement!

Multiple Query HSC States

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g)^{\otimes m} \rho_{\mathcal{H}}^{\otimes m} D_R(g^{-1})^{\otimes m}$$

Diagonal representation of the group

$$D_R(g)^{\otimes m} = \left[\bigoplus_{\mu} (I_{d_\mu} \otimes D_\mu(g)) \right]^{\otimes m}$$

Multiple Query HSC States

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g)^{\otimes m} \rho_{\mathcal{H}}^{\otimes m} D_R(g^{-1})^{\otimes m}$$

Diagonal representation of the group

$$D_R(g)^{\otimes m} = \left[\bigoplus_{\mu} (I_{d_\mu} \otimes D_\mu(g)) \right]^{\otimes m}$$

$$= \bigoplus_{\mu_1, \ldots, \mu_m} I_{d_{\mu_1}} \otimes \cdots I_{d_{\mu m}} \otimes D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g)$$

Clebsch-Gordan Decomposition

$$D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g) = \bigoplus_{\mu} \left(I_{n_\mu(\mu_1,\ldots,\mu_m)} \otimes D_\mu(g) \right)$$

Averaging over this representation,

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g)^{\otimes m} \rho_{\mathcal{H}}^{\otimes m} D_R(g^{-1})^{\otimes m}$$

implies

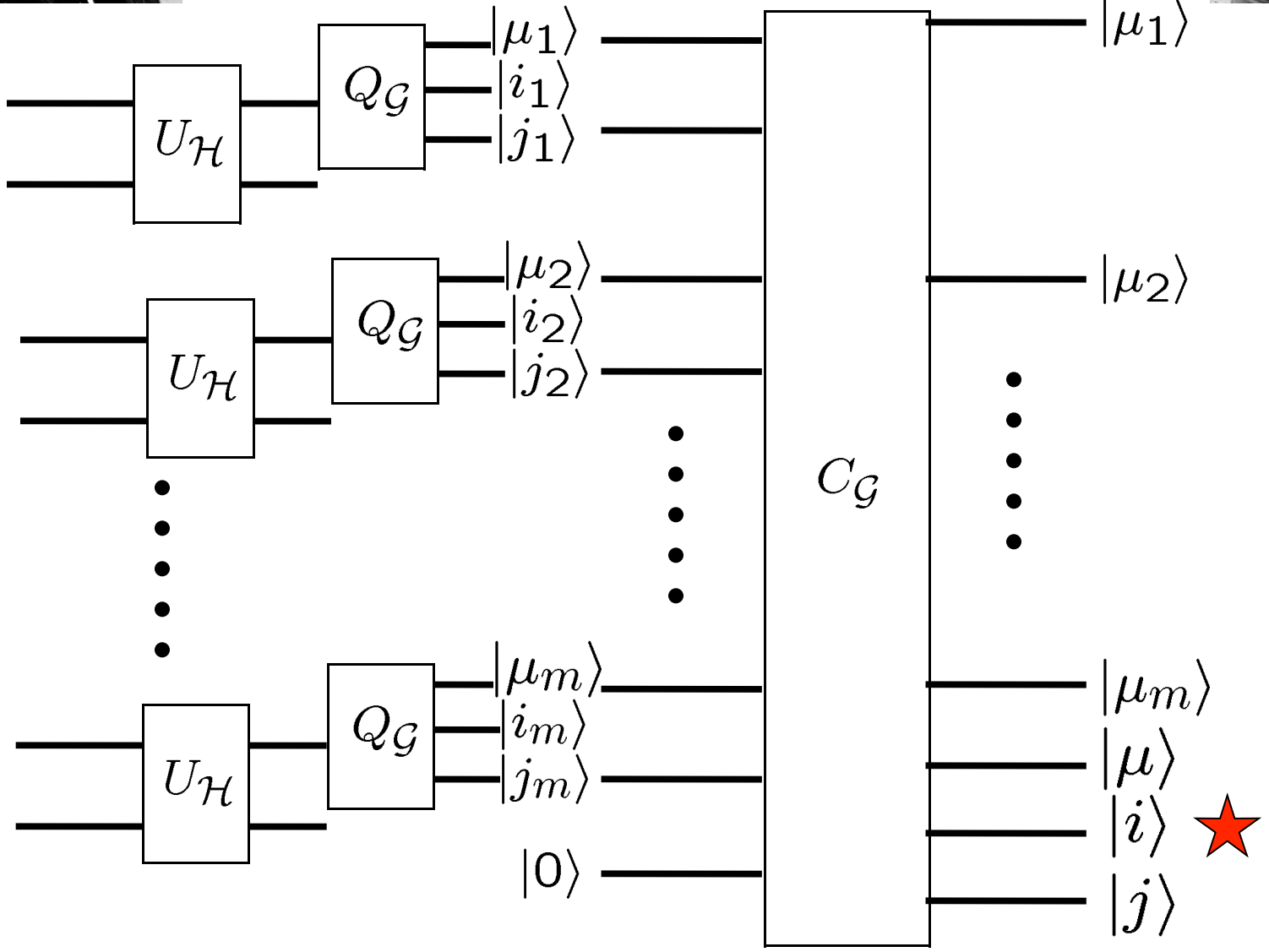
$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

$$D = d_{\mu_1} \cdots d_{\mu_m}$$



General Circuit





Quantum Algorithms for HSCP

1. Prepare *m* copies of the HS state.

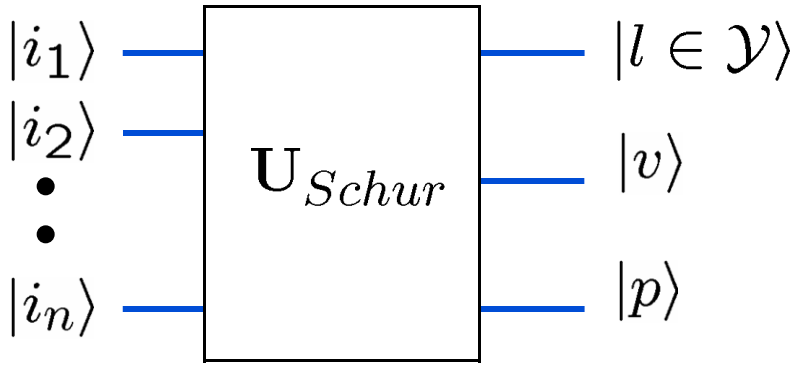
2. Perform the QFT over $\mathcal{G}$ on each copy.
- efficient circuits for most groups of interest
  [Moore, Russell, Rockmore 03]

3. Perform CG transform on resulting states
- efficient circuits?

4. Measurement on multiplicity index
- efficient circuits?

Schur Transform

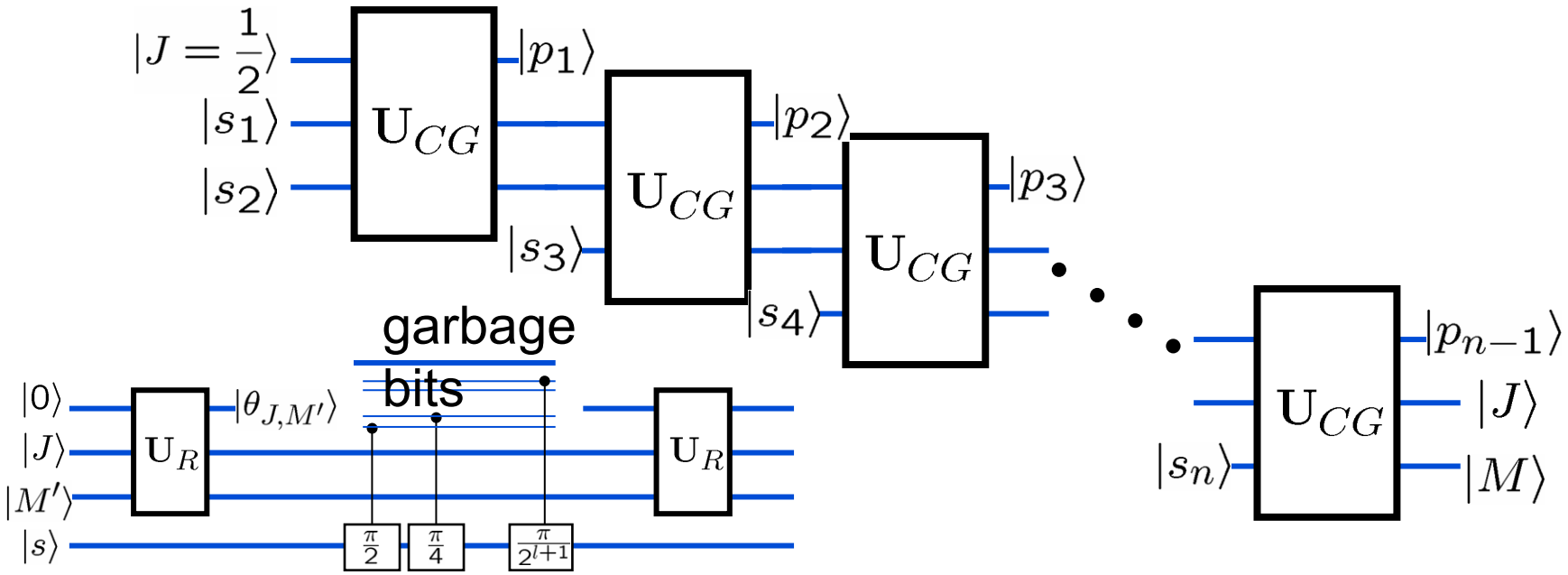
(work with Aram Harrow and Isaac Chaung)



$\mathcal{Y}$ = set of all partitions of n
(all Young Diagrams with n rows)

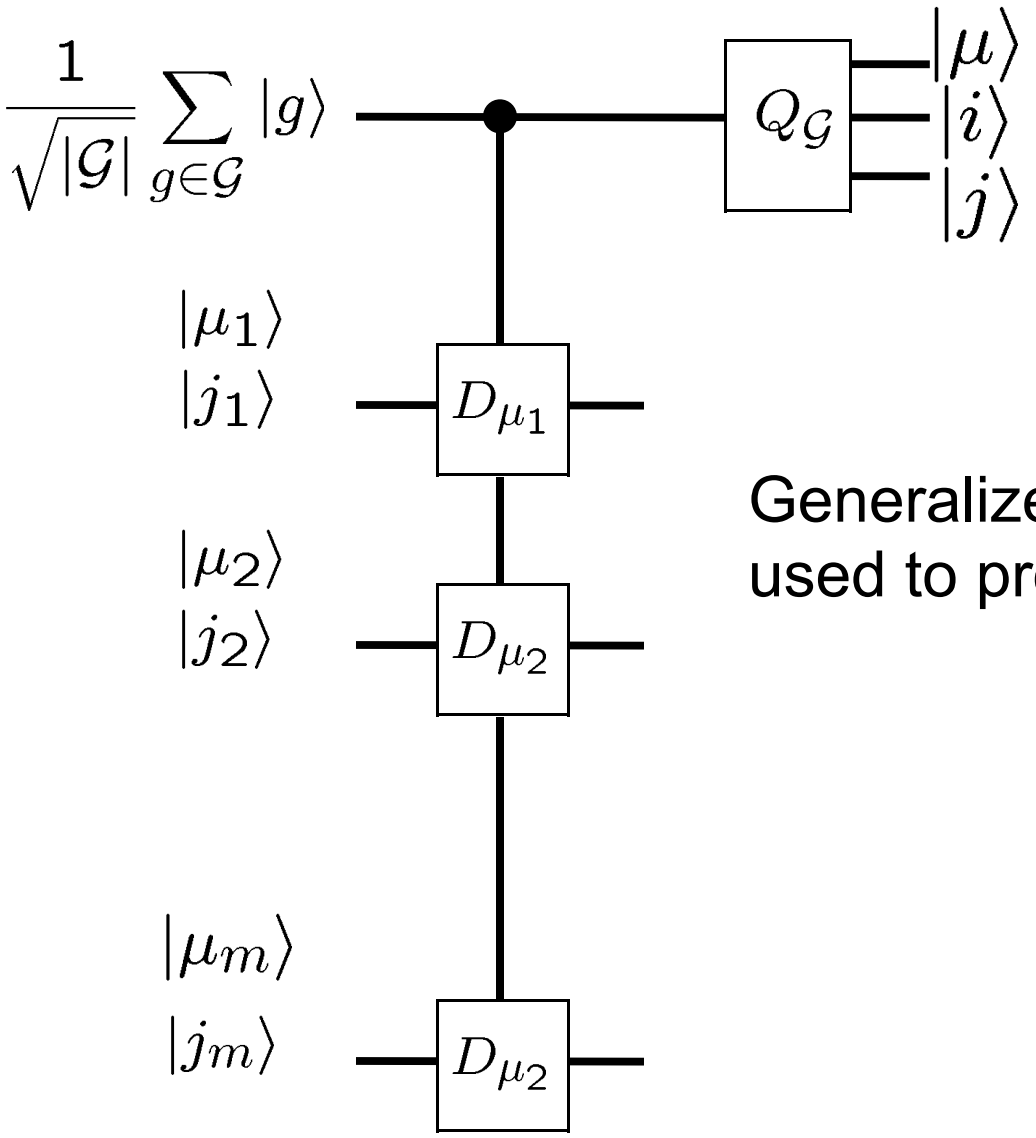
$$l = (l_1, l_2, \ldots, l_n) \qquad \sum_{i=1}^{n} l_i = n$$
$$l_1 \geq l_2 \geq \ldots \geq l_n \geq 0$$

Circuits of size polynomial in d,n,log(1/accuracy)



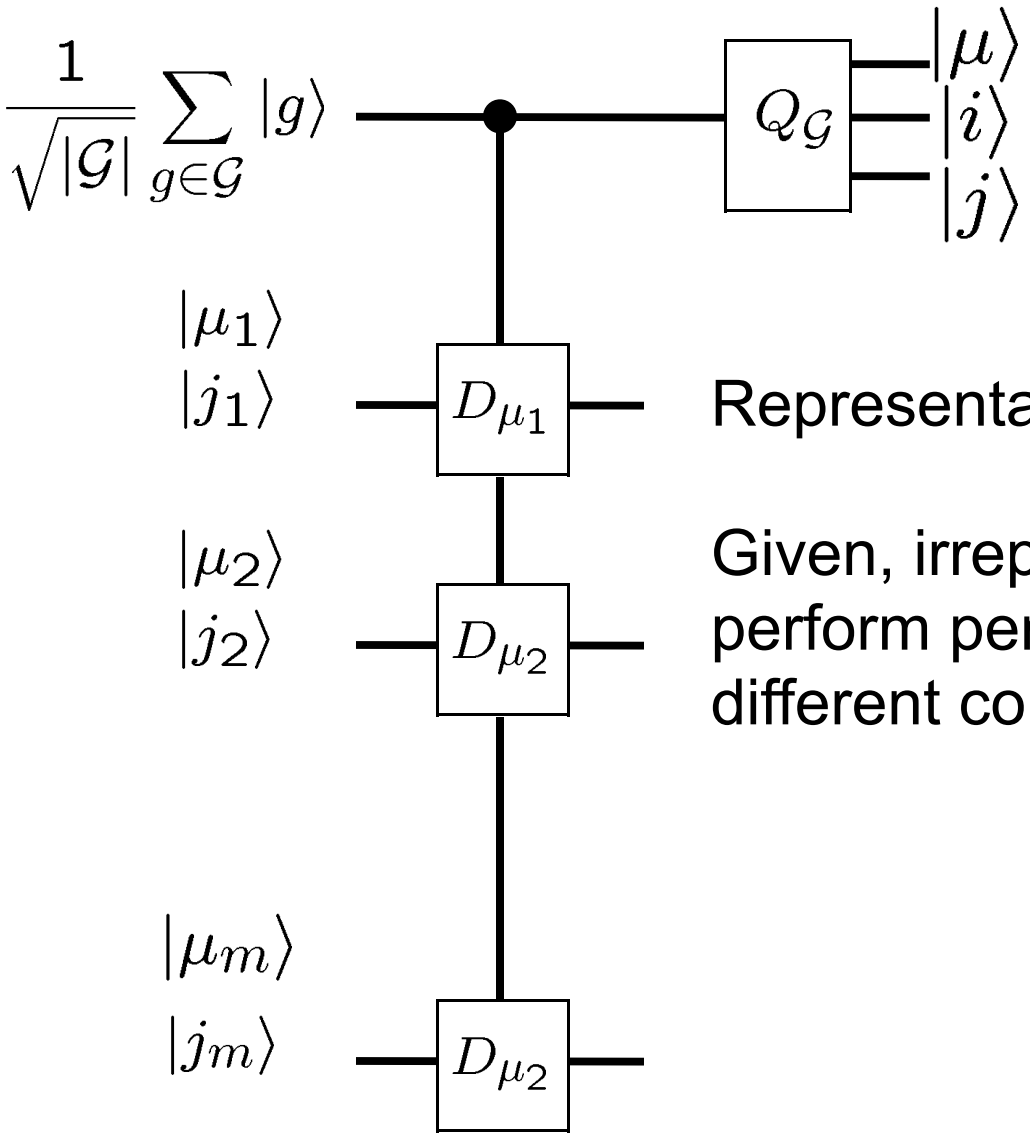
A Partial Solution





Generalized Kitaev circuit can be used to project onto CG subspaces

A Partial Solution



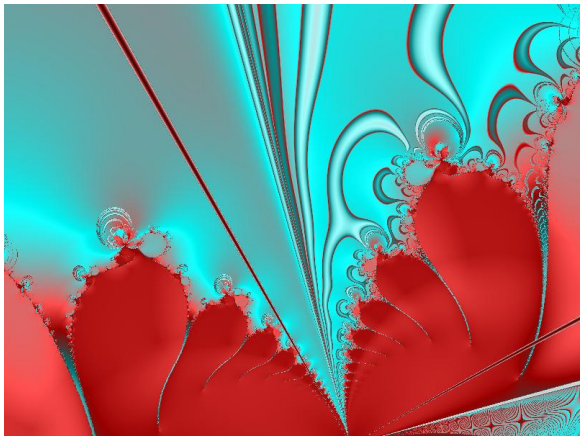
Representation theory problem:

Given, irreps, and sum in CG series, perform permutation between different consistent irreps.

What's My Point?

1. HSCP is (in the important instances at least)
   polynomial time equivalent to the HSP

2. In the same way that symmetry implies we should perform the QFT for the abelian case, the symmetry of the HSCP implies we should apply the Clebsch-Gordan transform over the finite group.



The Future

Efficient circuits for Clebsch-Gordan transforms over finite groups?

Identification of symmetries of the multiplicity space of
the Clebsch-Gordan transform over finite groups.



Dorit Aharonov's Question (Redux, Paraphrase)

How much of your life have you spent working on finding new algorithms for a quantum computer?

Never:
Hours:
Days:
Weeks:
Months:
Seasons:
Years:

Wasting Away In Hidden Subgroup Ville



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Racca

Department of Computer Science & Engineering

University of Washington